Fusing Global Structural and Local Deep Features for Thick Cloud Removal in Multitemporal Remote Sensing Images

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***Abstract*— Optical remote sensing images are inevitably affected by thick cloud cover, leading to information missing, which seriously hinders subsequent Earth observation tasks. Existing cloud removal methods typically focus on extracting either global or local features, but lack effective fusion of these two aspects, resulting in deficiencies in recovering global structure and fine details. However, fusing multiple features is challenging for single-driven methods due to the diversity of images and clouds. To this end, this article proposes a novel joint-driven cloud removal framework that can simultaneously exploit the global structural and local deep features of both the image and cloud components. Specifically, in our joint-driven scheme, we devise a flexible model-driven low-rank group sparse decomposition method, in which the global temporal–spectral correlations and shared sparse features of multitemporal remote sensing images and clouds from different scenes are effectively captured. To fully account for the differences in the local features of images and clouds from different scenes, we devise effective data-driven dual self-supervised networks to adaptively learn their local deep features. Additionally, we design an efficient half-quadratic splitting algorithm to iteratively optimize the joint-driven model, fostering mutual enhancement between the two and achieving further performance improvements. Experimental results show that the proposed method outperforms existing thick cloud removal approaches in both simulated and real-world datasets, under varying factors such as spectral bands, cloud coverage, time nodes, and spatial resolutions.**

***Index Terms*— Model driven, multitemporal remote sensing images, self-supervised network, thick cloud removal.**

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I. Introduction

With the development of remote sensing satellite technology, the quality of Earth observation data has significantly improved, particularly in temporal–spatial–spectral resolution. Such high-quality images are indispensable in various fields, including mineral exploration [1], image fusion [2], and urban planning [3]. However, studies indicate that the average annual cloud cover on Earth reaches 66% [4], which makes optical remote sensing imaging highly susceptible to cloud interference, significantly undermining image quality and affecting downstream applications. Consequently, the development of efficient methods for handling cloud cover in remote sensing images has become an urgent priority.

In recent years, cloud removal technology for remote sensing imagery has attracted significant attention, prompting the development of various methods to tackle this challenge. These approaches can generally be classified into two categories: model-driven optimization methods and data-driven deep learning methods. Model-driven optimization methods employ mathematical models designed to minimize cloud-induced distortion in images. These methods leverage physical priors to construct regularization terms, such as total variation (TV) [5], sparsity [6], and low-rankness [7], which help to constrain the solution space for ill-posed cloud removal tasks and effectively recover cloud-obscured image information. Despite their robust generalization, these methods are computationally intensive and heavily reliant on accurate model assumptions, hindering the capture of intricate, dynamic local features in image and cloud components.

With the rapid advancement of deep learning (DL), data-driven DL methods have also gained significant traction and demonstrated remarkable success [8], [9], [10], [11]. These methods automatically extract features and perform well on large-scale datasets. Furthermore, they can learn local deep features through an end-to-end process, enabling them to directly recover cloud-free images from cloudy ones [12], [13]. However, these approaches often require large paired datasets

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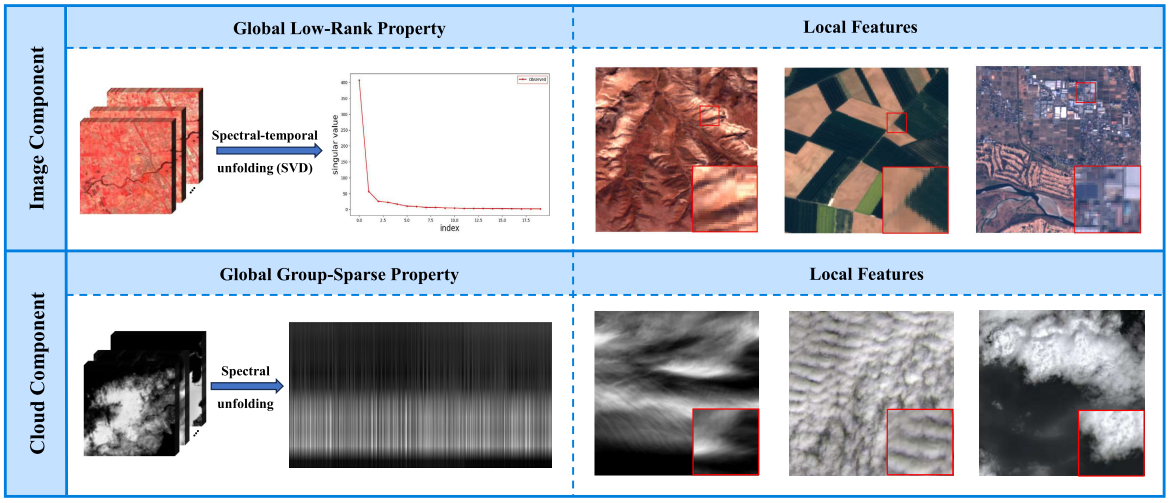


Fig. 1. Global and local features of clean image and cloud components. The left and right sides display the global and local features of the clean image and the cloud component, respectively. We can observe that the clean image and cloud component exhibit global low-rank and global group sparsity properties, respectively. Moreover, we can find that the local features vary greatly across different image and cloud scenes. For example, the agricultural scene is relatively smooth, while the village scene has more details.

for training, which makes them prone to overfitting, especially when applied to diverse real-world conditions. In addition, the appearance of clouds can vary greatly due to factors such as altitude, geography, and seasonal changes, posing challenges for existing methods that are unable to fully capture these complex global structural features. In general, existing methods can effectively exploit their respective advantages to capture the global structural or local features of images. However, due to the limitations of different types of methods, it is difficult to simultaneously and equally consider both the image and cloud components, and explore and fuse their respective features for cloud removal.

To tackle this challenge, we propose a novel joint-driven cloud removal framework that can simultaneously exploit and fuse the global structural and local deep features of both the image and cloud components, allowing the features of both the image and cloud to be fully revealed within this framework. We treat the observed image as a combination of a clean image and cloud components, allowing us to investigate their respective feature priors-global structural and local deep features (see Fig. 1). Multitemporal remote sensing images (MRSIs) capture the same surface area over time, exhibiting strong low-rank properties in the spectral–temporal dimension. We analyze the singular value distribution of the unfolded matrix of MRSIs in the spectral–temporal dimension. It can be observed that the singular values have obvious decaying trends, indicating the global low-rank property of images. Meanwhile, the cloud component appears in a specific region at a given time node, and its position changes over time, exhibiting a global group-sparse property in the spectral dimension. Data-driven deep learning methods struggle to adaptively learn such global structural features, while model-driven optimization methods are well-suited to design corresponding prior regularization terms to preserve these structures. Thus, we adopt a model-driven low-rank group sparsity decomposition framework to constrain the global structural features of image and cloud components. However, because of the lack of local feature representation, global regularization alone is insufficient for efficient image recovery. Local features vary significantly across different imaging scenes and times, and model-driven priors cannot fully capture these variations. By contrast, deep learning can adaptively learn these distributions. To address this, we design a dual self-supervised network to explore the local deep priors of both the image and cloud components. In summary, the global and local priors of the image and cloud components complement each other during iterative optimization, enabling better separation of the clean and cloud images. The illustration of the proposed method is shown in Fig. 2. The main contributions are summarized as follows.

1) We propose a novel joint-driven cloud removal framework that treats the clean image and cloud components as equally important targets. A hybrid modeling strategy combining model priors and deep networks is employed for both components to fully leverage the advantages of both model-driven and data-driven methods. To the best of our knowledge, this is the first approach to thoroughly consider the features of the cloud component and design a deep network to learn and estimate it.
2) We employ a flexible model-driven low-rank group sparse decomposition framework to characterize the shared global features of image and cloud components, and develop an efficient data-driven dual self-supervised network to adaptively learn their respective local details. This design enables effective integration of global and local information for both components.
3) We design an efficient half-quadratic splitting algorithm (HQS) to solve the proposed joint-driven framework. Experimental results illustrate that, under various datasets and factors such as spectral bands, cloud coverage, time points, and spatial resolution, the proposed method outperforms state-of-the-art thick cloud removal methods.

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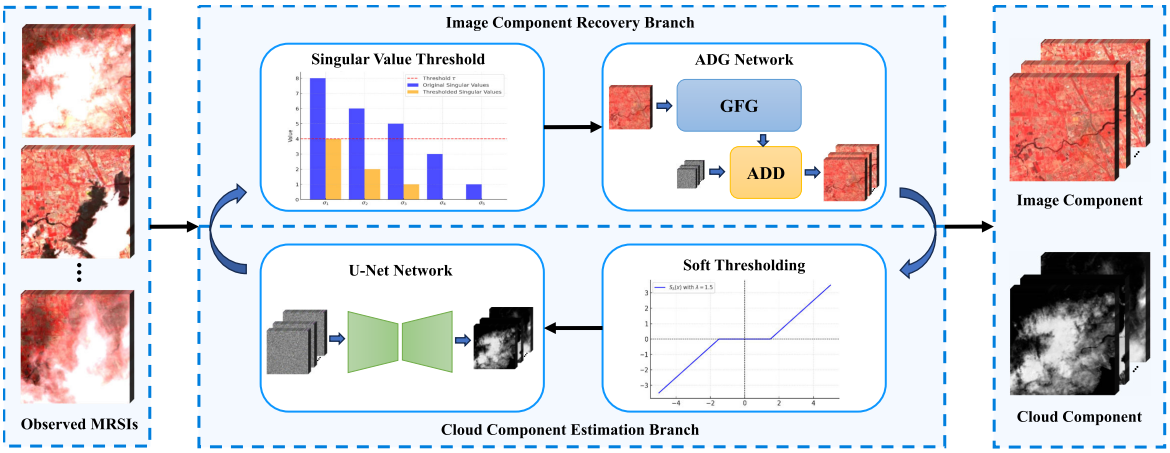


Fig. 2. Illustration of the proposed thick cloud removal method. We treat the clean image and cloud components equally and consider their global and local features, respectively. The global features are characterized using model-driven structural priors, while the local features are adaptively learned through a self-supervised deep network. The two components are iteratively estimated and mutually promoted under different constraints.

II. Related Works

In this section, we review the related work for cloud removal in remote sensing images. We roughly classify existing methods into two categories, model-driven optimization methods and data-driven deep learning-based methods.

A. Model-Driven Optimization Methods

Model-driven optimization methods significantly improve traditional cloud removal techniques based on inpainting and interpolation by incorporating prior information [7], [8], [14]. This type of method mainly includes nonblind and blind approaches, depending on whether a precise cloud mask needs to be provided. Nonblind methods treat cloud location information as missing data, regard the cloud removal problem as an image completion problem, and design low-rank completion models to reconstruct the cloud location information.

Early methods achieved this by employing low-rank matrix completion techniques [15], [16]. However, given that MRSIs are high-dimensional tensors, matrix-based methods face limitations in capturing global spatial–spectral–temporal correlation. To address this, low-rank tensor completion methods [17], [18], [19], [20] were introduced, taking advantage of their multidimensional data processing capabilities to better capture global correlations and complex cloud structures. He et al. [21] combined tensor ring completion with total variation optimization to achieve remote sensing image reconstruction. Chen et al. [22] further enhanced low-rank joint spatiotemporal constraints by integrating frequency spectrum modulation into the tensor completion framework. However, when the given mask is inaccurate, achieving satisfactory cloud removal results remains a challenging task. To reduce the reliance on masks, Li et al. [16] proposed a semi-blind method based on tensor decomposition, which improves the model's robustness by providing inaccurate masks. Chen et al. [5] first proposed a blind cloud removal method based on low-rank sparse decomposition, detecting masks from the estimated cloud components. To further exploit the deep prior of the image, Chen et al. [23] and Zhang et al. [24] integrated image models and depth priors to remove thick clouds from multitemporal images. Despite these advancements, such methods still face challenges, including limited adaptability to complex cloud shapes and reliance on model assumptions.

B. Data-Driven Deep Learning Methods

With the rapid development of deep learning techniques, data-driven deep learning methods have gained increasing attention for cloud removal in recent years [25], [26], [27], employing their local connectivity and weight-sharing properties to effectively capture hierarchical spatial features of the image. These methods leverage the capacity of neural networks to automatically learn features, overcoming the limitations of purely model optimization approaches in feature extraction. First, convolutional neural network (CNN) approaches are widely employed to learn the nonlinear feature in cloud removal tasks [9], [12], [28]. Zhang et al. [29] proposed a progressive spatial–temporal patch group learning framework for cloud removal. Jiang et al. [30], [31] incorporated frequency-domain attention into the CNN to further enhance feature extraction, aiming at the removal of thick and thin clouds, respectively. Li et al. [32] combined multimodal data with CNNs for cloud removal in single-temporal remote sensing images. Then, generative adversarial networks (GANs) have also been applied to cloud removal, which can learn the underlying distribution of the data through the adversarial process between the generator and the discriminator [13], [33], [34]. More recently, diffusion model-based methods have gained popularity in cloud removal problems [35]. Zhao and Jia [36] presented a serial diffusion model, which significantly reduces the time cost of model retraining. Although supervised deep learning methods perform well and offer fast processing when test data shares similar characteristics with the training data, the lack of paired datasets in cloud removal tasks limits their ability to generalize across different scenarios. Furthermore, DL-based methods are often inadequate in capturing global structural features of images and tend to overlook cloud components, which are equally important as the image content itself.

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III. Methodology

A. Problem Formulation

The primary goal of multitemporal remote sensing cloud removal is to reconstruct a cloud-free image $\mathcal{X}$ from cloud-contaminated observation $\mathcal{Y}$. To better preserve the information of cloud-free regions, we assume the degradation model of thick cloud contamination as follows:

$$\mathcal{Y} = \mathcal{M} \odot \mathcal{X} + (1 - \mathcal{M}) \odot \mathcal{C} \tag{1}$$

where $\mathcal{Y} \in \mathbb{R}^{m\times n\times b\times t}$, $\mathcal{X} \in \mathbb{R}^{m\times n\times b\times t}$, and $\mathcal{C} \in \mathbb{R}^{m\times n\times b\times t}$ are cloud-contaminated image, clean (cloud-free) image, and cloud component, respectively. m, n, b, and t correspond to the image dimensions: height, width, spectral band, and time node, respectively. $\mathcal{M} \in \mathbb{R}^{m\times n\times b\times t}$ is a binary mask where zeros indicate thick cloud-contaminated regions and ones indicate cloud-free regions.

Due to the reconstruction problem being an ill-posed inverse problem, it is difficult to recover $\mathcal{X}$ directly from the degradation model (1). Model-driven optimization methods design suitable regularization terms to restrict the solution space, and the reconstruction model can be formulated as

$$\min_{\mathcal{X},\mathcal{C}} \frac{1}{2}\|\mathcal{Y} - \mathcal{M} \odot \mathcal{X} - (1 - \mathcal{M}) \odot \mathcal{C}\|_F^2 + \lambda_1 R(\mathcal{X}) + \lambda_2 R(\mathcal{C}) \tag{2}$$

where the first term is data-fidelity term, $R(\mathcal{X})$ and $R(\mathcal{C})$ represent regularization terms encoding prior information on $\mathcal{X}$ and $\mathcal{C}$, respectively, and λ_1 and λ_2 are regularization parameters balancing these two terms. Although handcrafted priors designed by such methods can effectively capture the global structural features of image and cloud components, they overlook their local features, resulting in poor reconstruction performance. Data-driven deep learning methods train an end-to-end network from observed images to clean images using large datasets, i.e.,

$$\mathcal{X} = \mathcal{S}_\theta(\mathcal{Y}) \tag{3}$$

where $\mathcal{S}_\theta(\cdot)$ is the end-to-end network, which can effectively learn the local features of the image, and θ is the network parameter. When the test data share the same characteristics as the training data, these methods achieve high reconstruction accuracy and fast speed. However, in real-world scenarios, fully paired training datasets are unavailable, and it is difficult to preserve the global features shared across different scenes. In addition, these methods overlook the cloud components, which are equally important as image information. Thus, how to simultaneously and equally consider both the image and cloud components, and effectively fuse their respective global and local features, is a challenge in the cloud removal task.

B. Proposed Joint-Driven Model

1) Global Structural Features of Image and Cloud Components: Global structural features refer to the shared characteristics present across different data, which help in understanding the overall structure of both cloud and noncloud regions, thus improving image reconstruction. Fully exploring these shared features not only enhances the flexibility of reconstruction but also ensures that the reconstructed results do not deviate from the real image.

a) Global structural feature of image component: In MRSIs, spectral features of the same scene at different time nodes exhibit strong global correlations because of the consistency of the imaging target, with variations occurring solely in the temporal dimension due to changes in lighting and atmospheric conditions. This inherent high correlation is physically manifested as the spectral–temporal low-rank property of the image component [5]. Fig. 1 presents the singular value distribution of the unfolded matrix of MRSIs in the spectral–temporal dimension. We can find that the singular values have obvious decaying trends, indicating the global low-rank property of images. In contrast, due to the dynamic nature of cloud distributions at different time nodes, the cloud component typically exhibits minimal correlation along the temporal dimension, making it easier to separate the image and cloud components.

b) Global structural feature of cloud component: With clouds affecting only a small proportion of the entire image sequence in MRSIs, cloud coverage is typically sparse across the temporal dimension. This property naturally lends itself to sparse prior modeling, and a common approach is to employ the l_1 norm. However, while the l_1 norm effectively captures global sparsity, it fails to account for the correlation between the spectral band. In MRSIs, the thick cloud component generally exhibits spectral consistency, i.e., the spectral reflectance of clouds remains consistent across different bands, implying that if a pixel is cloud-covered in one band, it is highly likely to be cloud-covered in others. Fig. 1 shows a schematic of the cloud component at each time point unfolded along the spectral dimension. It can be seen that the number of nonzero columns is significantly smaller than the number of zero columns, which indicates the global group sparse structure of the cloud component along the spectral dimension. In contrast, this global structural feature is not possessed by the image components but is unique to the cloud components.

Data-driven deep learning methods can effectively learn the local features of different data, but they struggle to simultaneously capture the global structural features of both the image and cloud components. On the other hand, model-driven methods are capable of capturing these global structural features. Therefore, we adopt a low-rank group sparsity framework [7], [23] to characterize the global structural features of both the image and cloud components for thick cloud removal, i.e.,

$$\min_{\mathcal{X},\mathcal{C}} \frac{1}{2}\|\mathcal{Y} - \mathcal{M} \odot \mathcal{X} - (1 - \mathcal{M}) \odot \mathcal{C}\|_F^2 + \lambda_1\|\mathcal{X}\|_* + \lambda_2\|\mathcal{C}\|_{2,1}. \tag{4}$$

To ensure the consistency of variables, we define $\|\mathcal{X}\|_*$ as $\sum_i \sigma_i(\mathcal{R}(\mathcal{X}))$, where $\mathcal{R}$ is a reshape operator, i.e., $\mathcal{R}$: $\mathcal{X} \in \mathbb{R}^{m\times n\times b\times t} \to \mathbf{X} \in \mathbb{R}^{mn\times bt}$, and $\sigma_i(\mathbf{X})$ is the i-th singular value of $\mathbf{X}$. Additionally, the $\ell_{2,1}$-norm of the tensor $\mathcal{C}$ is given by $\|\mathcal{C}\|_{2,1} = \sum_i \sum_j \sum_q \|\mathcal{C}(i, j, :, q)\|_2$.

2) Local Deep Features of Image and Cloud Components: The model-driven low-rank group sparsity model (4) can preserve the global feature of reconstructed images. However, because of the lack of exploration of local features, global

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feature regularization alone cannot reconstruct the MRSIs efficiently. Model-driven methods typically use total variation regularization to capture local features, but these handcrafted priors often do not fit all scenarios. In remote sensing images, because of differences in imaging areas, the local feature variations between different data are quite large. Fig. 1 shows the local features of three different scenes: mountain, farmland, and village areas. We can observe that the local features of farmland are relatively smooth, while the local features of mountain and village areas contain more details. Similarly, the local features are highly variable for cloud components.

The highly variable local features make it difficult for model-driven methods to accurately capture them using handcrafted priors. However, data-driven deep learning methods compensate for this limitation, as they can adaptively learn the local features of the image, thereby better reconstructing the image details. Previous data-driven deep learning methods require a large training dataset to reconstruct images, but such datasets are difficult to obtain for cloud removal tasks. To avoid relying on training datasets, we propose a novel dual self-supervised network to extract the local deep features of both the image and cloud components, i.e.,

$$\mathcal{X} = \mathcal{F}_\Theta(\mathcal{E}_1, \mathcal{G}), \quad \mathcal{C} = \mathcal{T}_\Omega(\mathcal{E}_2) \tag{5}$$

where $\mathcal{F}_\Theta(\cdot)$ and $\mathcal{T}_\Omega(\cdot)$ are two self-supervised networks, Θ and Ω are network's parameters to be learned, $\mathcal{E}_1$ and $\mathcal{E}_2$ are random noise inputs to the network, and $\mathcal{G}$ is the guidance image input to the network $\mathcal{F}_\Theta(\cdot)$.

3) Proposed Thick Cloud Removal Model: By fusing the global structural and local deep features of both the image and cloud components, and substituting (5) into (4), we obtain the proposed joint-driven thick cloud removal model as follows:

$$\begin{aligned}\min_{\Theta,\Omega} \frac{1}{2}\|\mathcal{Y} - \mathcal{M} \odot \mathcal{F}_\Theta(\mathcal{E}_1, \mathcal{G}) - (1-\mathcal{M}) \odot \mathcal{T}_\Omega(\mathcal{E}_2)\|_F^2 \\ + \lambda_1 \|\mathcal{F}_\Theta(\mathcal{E}_1, \mathcal{G})\|_* + \lambda_2 \|\mathcal{T}_\Omega(\mathcal{E}_2)\|_{2,1}.\end{aligned} \tag{6}$$

The proposed model (6) equally considers both the image and cloud components, and separately extracts their global and local features, ensuring that both image and cloud features are fully captured. Additionally, the proposed model integrates the advantages of model-driven and data-driven methods, allowing these two approaches to complement each other. As a result, the model enables effective separation of the image and cloud components, leading to superior and flexible cloud removal.

C. Optimization Algorithm

To solve the minimization problem (6), we design an effective HQS algorithm. By introducing two auxiliary variables $\mathcal{W}$ and $\mathcal{Z}$, we rewrite the problem (6) as

$$\begin{aligned}\min_{\Theta,\Omega,\mathcal{W},\mathcal{Z}} \frac{1}{2}\|\mathcal{Y} - \mathcal{M} \odot \mathcal{F}_\Theta(\mathcal{E}_1, \mathcal{G}) - (1-\mathcal{M}) \odot \mathcal{T}_\Omega(\mathcal{E}_2)\|_F^2 \\ + \lambda_1 \|\mathcal{W}\|_* + \lambda_2 \|\mathcal{Z}\|_{2,1} \\ \text{s.t. } \mathcal{W} = \mathcal{F}_\Theta(\mathcal{E}_1, \mathcal{G}), \quad \mathcal{Z} = \mathcal{T}_\Omega(\mathcal{E}_2).\end{aligned} \tag{7}$$

The constrained problem above can be solved by minimizing the following penalty function:

$$\begin{aligned}\min_{\Theta,\Omega,\mathcal{M},\mathcal{W},\mathcal{Z}} \frac{1}{2}\|\mathcal{Y} - \mathcal{M} \odot \mathcal{F}_\Theta(\mathcal{E}_1, \mathcal{G}) - (1-\mathcal{M}) \odot \mathcal{T}_\Omega(\mathcal{E}_2)\|_F^2 \\ + \lambda_1 \|\mathcal{W}\|_* + \lambda_2 \|\mathcal{Z}\|_{2,1} + \frac{\rho_1}{2}\|\mathcal{F}_\Theta(\mathcal{E}_1, \mathcal{G}) - \mathcal{W}\|_F^2 \\ + \frac{\rho_2}{2}\|\mathcal{T}_\Omega(\mathcal{E}_2) - \mathcal{Z}\|_F^2\end{aligned} \tag{8}$$

where ρ_1 and ρ_2 are penalty parameters. By applying the basic framework of HQS, we decompose the solution of the above objective function into the iterative optimization of the following subproblems.

1) Update Θ: The Θ-subproblem is given as follows:

$$\begin{aligned}\arg\min_{\Theta} \frac{1}{2}\|\mathcal{Y} - \mathcal{M} \odot \mathcal{F}_\Theta(\mathcal{E}_1, \mathcal{G}) - (1-\mathcal{M}) \odot \mathcal{T}_\Omega(\mathcal{E}_2)\|_F^2 \\ + \frac{\rho_1}{2}\|\mathcal{F}_\Theta(\mathcal{E}_1, \mathcal{G}) - \mathcal{W}\|_F^2\end{aligned} \tag{9}$$

which is a quadratic minimization problem. We employ the Adam algorithm [37] as the optimizer to update the network parameter Θ in our proposed self-supervised attention-based deep guidance network (see Section III-D). The loss function is defined as (9), which can denoise $\mathcal{W}$ and minimize the data fidelity term. Note that the proposed self-supervised network does not require an external training dataset for pretraining and the network parameter Θ is randomly initialized. Due to the continuous changes in clouds, cloud-free data can be collected during cloud-free periods in MRSIs. To make full use of these cloud-free data, the guidance image $\mathcal{G}$ is set as the average of cloud-free images at different time nodes.

2) Update $\mathcal{W}$: The $\mathcal{W}$-subproblem is given as follows:

$$\arg\min_{\mathcal{W}} \lambda_1 \|\mathcal{W}\|_* + \frac{\rho_1}{2}\|\mathcal{F}_\Theta(\mathcal{E}_1, \mathcal{G}) - \mathcal{W}\|_F^2 \tag{10}$$

and the closed-form solution can be efficiently obtained by using the singular value thresholding (SVT) method [38], as shown below:

$$\mathcal{W} = \mathcal{R}^{-1}\left(U \Sigma_+ V^T\right) \tag{11}$$

where $\Sigma_+ = \max(\Sigma - \lambda_1/\rho_1, 0)$, $U \Sigma V^T = \mathcal{R}(\mathcal{F}_\Theta(\mathcal{E}_1, \mathcal{G}))$, $\mathcal{R}$ is the unfolding operator from tensor to matrix, and $\mathcal{R}^{-1}$ is the folding operator from matrix to tensor.

3) Update Ω: The Ω-subproblem is given as follows:

$$\begin{aligned}\arg\min_{\Omega} \frac{1}{2}\|\mathcal{Y} - \mathcal{M} \odot \mathcal{F}_\Theta(\mathcal{E}_1, \mathcal{G}) - (1-\mathcal{M}) \odot \mathcal{T}_\Omega(\mathcal{E}_2)\|_F^2 \\ + \frac{\rho_2}{2}\|\mathcal{T}_\Omega(\mathcal{E}_2) - \mathcal{Z}\|_F^2.\end{aligned} \tag{12}$$

Since the local features of cloud components are simpler than those of image components, we use a self-supervised U-Net [39] to learn the local deep features of cloud components. Similarly, Adam is used as the optimizer for the U-Net, and problem (12) is used as the loss function.

4) Update $\mathcal{Z}$: The $\mathcal{Z}$-subproblem is given as follows:

$$\arg\min_{\mathcal{Z}} \lambda_2 \|\mathcal{Z}\|_{2,1} + \frac{\rho_2}{2}\|\mathcal{T}_\Omega(\mathcal{E}_2) - \mathcal{Z}\|_F^2. \tag{13}$$

We set $\mathcal{B} = \mathcal{T}_\Omega(\mathcal{E}_2)$, and the closed-form solution of $\mathcal{Z}$ is achieved by employing the soft-shrinkage operator

$$\mathcal{Z}_{i,j,:,q} = \begin{cases} \dfrac{\|\mathcal{B}_{i,j,:,q}\|_2 - \lambda_2/\rho_2}{\|\mathcal{B}_{i,j,:,q}\|_2} \mathcal{B}_{i,j,:,q}, & \text{if } \|\mathcal{B}_{i,j,:,q}\|_2 > \dfrac{\lambda_2}{\rho_2} \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

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5) Update $\mathcal{M}$: To ensure cloud removal results are not dependent on a predefined cloud mask, we adaptively update it. Since thick clouds and land cover objects (e.g., water bodies and vegetation) exhibit significant differences in their spectral characteristics [40], thick clouds typically show higher reflectance in the visible and near-infrared bands. Therefore, by setting appropriate thresholds, it is possible to effectively and conveniently distinguish clouds from other land cover types. Consequently, we designed the following algorithm to obtain a thick cloud mask for cloud locations:

$$\mathcal{M}_{i,j,:,q} = \begin{cases} 1, & \text{if } \frac{1}{b}\sum_{p=1}^{b}\left|\mathcal{C}_{i,j,p,q}\right| < \epsilon \\ 0, & \text{otherwise} \end{cases} \tag{15}$$

where $\epsilon > 0$ denotes a given threshold value. To retain information in cloud-free regions, the final reconstruction result is achieved by

$$\mathcal{X} = \mathcal{M} \odot \mathcal{Y} + (1 - \mathcal{M}) \odot \mathcal{F}_{\Theta}(\mathcal{E}_1, \mathcal{G}). \tag{16}$$

Summarizing the optimization procedure of the whole process, we obtain the pseudocode of the proposed method for cloud removal in Algorithm 1.

Algorithm 1 Proposed Thick Cloud Removal Method

Require: $\mathcal{Y}$, λ_1, λ_2 and ϵ.

Initialize: $\mathcal{X} = \mathcal{Y}$, $\mathcal{W} = \mathcal{Y}$, $\mathcal{Z} = \mathcal{Y}$, $\mathcal{M} = \mathcal{O}$, $\rho_1 = 0.0001$, $\rho_2 = 0.05$, $k = 0$, $k_{max} = 30$.
while not convergence **do**
  Update Θ by ADG with loss function in Eq. (9)
  Update $\mathcal{W}$ by Eq.(11);
  Update Ω by U-Net with loss function in Eq.(12);
  Update $\mathcal{Z}$ by Eq.(14);
  Update $\mathcal{M}$ by Eq. (15);
  Update penalty parameters by $\rho_1 = 1.02\rho_1$ and $\rho_2 = 1.02\rho_2$
  Update iterations by $k = k + 1$.
  Update $\mathcal{X}$ by Eq. (16).
  Check the convergence condition: $\frac{\|\mathcal{X}^{k+1} - \mathcal{X}^k\|_F}{\|\mathcal{X}^k\|_F} \leq 10^{-3}$ and $k < k_{max}$.
end while

Ensure: Cloud removal image $\mathcal{X}$.

D. Attention-Based Deep Guidance Network

Physical prior effectively explores the intrinsic global characteristic of surface information, but it fails to capture more complex and deeper features. To better preserve the complex local features of the image and fully utilize the high-quality images from cloud-free time nodes, we propose an attention-based deep guidance (ADG) self-supervised network. Inspired by previous works [41], [42], we design the ADG shown in Fig. 3, which comprises two main components: the guided feature generator (GFG) and the attention-based deep decoder (ADD). The GFG generates multiscale guided features from the guidance image, facilitating the parameter optimization and image generation of ADD.

1) Guided Feature Generator: As shown in Fig. 3 (a), GFG adopts an encoder–decoder architecture with a skip connection to extract multiscale features from the guidance image. The process begins with a convolution operation applied to the guidance image $\mathcal{G}$, producing the initial input feature E_0. This feature is subsequently processed through a series of encoding blocks, and E_k denote the features of the encoder at the k-th scale ($k = 0, 1, \ldots, N$). Each encoding block involves operations such as downsampling, convolution, and convolutional block attention module (CBAM) [43]. CBAM is a lightweight attention mechanism designed to refine feature representations by sequentially applying channel and spatial attention. It adaptively highlights important features while suppressing less relevant ones, enhancing the network's ability to capture features across both channel and spatial dimensions.

Similarly, the decoding block also consists of a convolution operation and a CBAM module with detailed operations shown in Fig. 3(d). The feature D_1 entering the decoder is obtained by upsampling the output E_N of the final encoding layer, where D_k denotes the k-th decoding layer feature ($k = 1, \ldots, N$). These encoding and decoding layer features are used as guidance information in ADD to assist in the recovery of image components. In this architecture, downsampling is implemented using 3×3 convolutions with a stride of 2, while upsampling is performed via bilinear interpolation to restore spatial features. The learning rate is set to 0.002.

2) Attention-Based Deep Decoder: Inspired by the design in [42], ADD is composed of two primary modules: the semantic attention module (SAM) and the detail refinement module (DRM). The SAM employs a gating mechanism to enhance semantic alignment between the output image and the guidance image. We denote the features of the deep decoder as F, which are initially derived from a random tensor $\mathcal{E}_1$ input into the deep decoder. The operational details of SAM are illustrated in Fig. 3(c) and can be expressed mathematically as follows:

$$\text{SAM}(F|D_k) = F \otimes D_k \tag{17}$$

where $\otimes$ represents the element-wise multiplication. It should be noted that the first layer of the SAM takes $\mathcal{E}_1$ and E_N as inputs, respectively.

In the GFG sub-network, the spatial resolution of low-level feature maps is diminished due to convolution and pooling operations, resulting in a loss of spatial detail information. The DRM mitigates this issue by weighting the deep spatial features with high-level semantic features from the decoder, effectively integrating semantic and spatial information. To avoid the suppression of spatial details in deep features caused by the multiplicative operations of traditional attention mechanisms, DRM employs additive weighting, which more effectively preserves fine-grained spatial information and enhances the integrity of feature representations. This mechanism leverages the spatial details extracted from the decoder to refine the deep features, aligning them more closely with the spatial structure of the original remote sensing image. The detailed operations are illustrated in Fig. 3(b), and this

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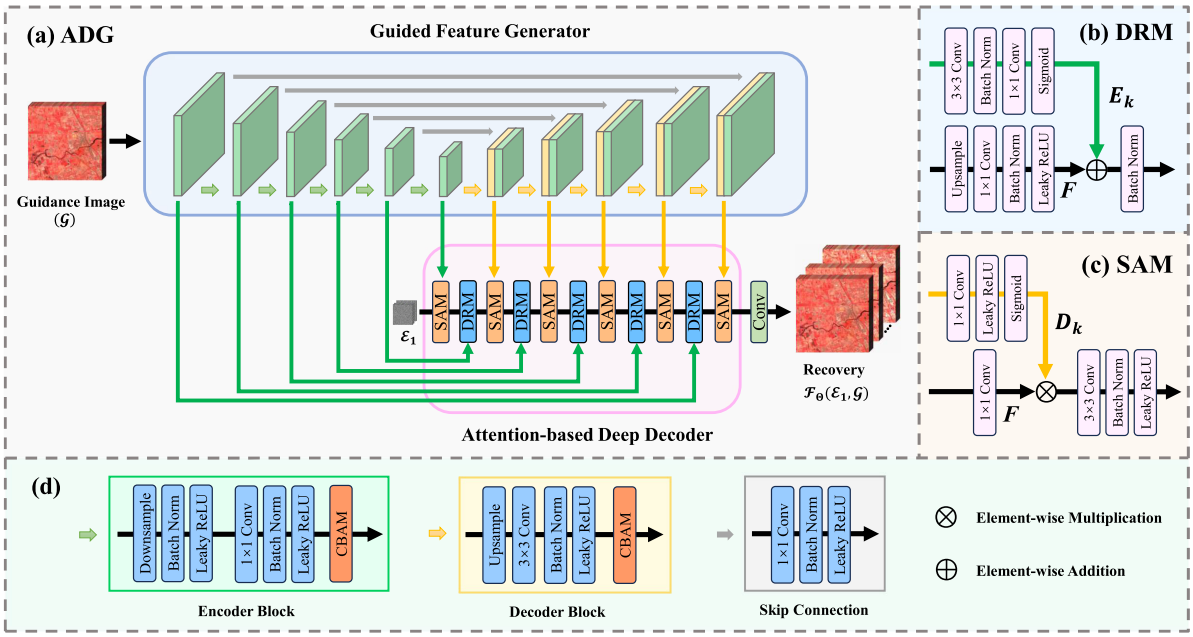


Fig. 3. Diagram of the ADG network. First, the guidance image is fed into GFG to extract its multiscale semantic features. Then, ADD utilizes these semantic features to guide parameter estimation and image generation in the depth decoder, where SAM is used for feature refinement and semantic alignment, and DRM supplements spatial information. The yellow and green short arrows represent the encoder and decoder blocks, respectively; the gray long arrows indicate skip connections; the other colored long arrows show data flow directions, and the cubes represent features at different scales. (a) ADG. (b) DRM. (c) SAM. (d) Details of the encoder blocks, decoder blocks, and skip connections.

process can be mathematically expressed as follows:

$$\mathrm{DRM}(F|E_k) = F \oplus E_k \tag{18}$$

where $\oplus$ represents the element-wise addition, and the dimensions of F and E_k are the same at each scale.

The designed ADG is capable of learning the complex local deep features of image components and can fully utilize the clean image from cloud-free time nodes. Additionally, since the network does not require pretraining on external datasets, it is optimized in a self-supervised manner, leveraging the advantages of model-driven approaches while benefiting from the efficiency of CNNs.

IV. Experiments

To evaluate the effectiveness of our proposed method, we conduct comparative experiments with six state-of-the-art thick cloud removal approaches, including traditional image inpainting-based method mosaicing [44], model-driven optimization methods such as total variation regularized low-rank sparse decomposition method (TVLRSDC) [5], robust thick cloud removal method (RTCR) [45], and self-supervised network-based method MT [9]; and two hybrid methods combining model and deep learning: spectral–temporal low-rank regularization with deep prior (STLR-DP) [46] and low-rank regularized self-supervised network (LRRSSN) [23]. Following the authors' guidelines and code configurations, the parameters for each method are tuned to their optimal settings to ensure fair comparisons. All experiments are executed on Inter i9-13900K, NVIDIA GeForce RTX 4090, and 64 GB of RAM. TVLRSDC and RTCR methods are implemented in MATLAB R2018a, while the other methods are implemented using PyTorch 2.1.2.

A. Simulated Experiment

1) Simulated Setting: To validate the flexibility and applicability of the proposed method in different types of sensors and scenes, we select two different scenes from the SEN12MS-CR-TS dataset[1] (farmland and mountain scenes) and different scenes from the Sentinel-2 dataset[2] (village and port scenes), resulting in four distinct test cases. Moreover, to investigate the robustness of the proposed method in various factors, four factors are considered in the experiments, namely spectral bands, cloud coverages, time nodes, and spatial resolutions. These factors are applied in the four scenes, respectively, with the specific settings detailed below. The design of the simulated experiment is based on DP-LRTSVD [24].

a) Farmland dataset: Since the farmland dataset exhibits unique spectral features across different spectral bands [47], which are often significantly influenced by the number of bands in MRSIs. To assess the robustness of the proposed method in terms of the number of bands, we consider cloud-contaminated images that contain different spectral bands, meaning the data dimension is $256 \times 256 \times b \times 5$ (where b = 4, 8, 12).

b) Mountain dataset: To test the performance of the proposed method under different cloud cover ratios, we set up three cases with varying levels of cloud coverage in the mountain dataset. The dimension of the test images is $256 \times 256 \times 4 \times 5$, with the cloud coverage areas set to represent 15%–30%, 30%–45%, and 45%–60% of the entire image, respectively.

[1]https://github.com/PatrickTUM/SEN12MS-CR-TS
[2]https://scihub.copernicus.eu/

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TABLE I
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE FARMLAND DATASET WITH VARYING NUMBERS OF BANDS

Spectral band	Index	Observed	TVLRSDC	Mosaicing	MT	RTCR	STLR-DP	LRRSSN	Ours
4	PSNR	8.485	30.874	24.808	32.839	29.214	35.717	34.844	**37.980**
	SSIM	0.762	0.933	0.895	0.910	0.923	0.963	0.952	**0.972**
	CC	0.192	0.831	0.555	0.900	0.779	0.970	0.948	**0.974**
8	PSNR	9.241	28.200	26.089	30.812	29.339	36.414	33.534	**37.267**
	SSIM	0.767	0.920	0.902	0.895	0.922	0.963	0.945	**0.966**
	CC	0.263	0.791	0.699	0.884	0.838	0.971	0.940	**0.975**
12	PSNR	9.597	31.926	28.640	31.811	32.759	38.153	38.047	**40.272**
	SSIM	0.776	0.941	0.917	0.906	0.933	0.969	0.960	**0.974**
	CC	0.239	0.781	0.656	0.773	0.795	0.914	0.947	**0.949**

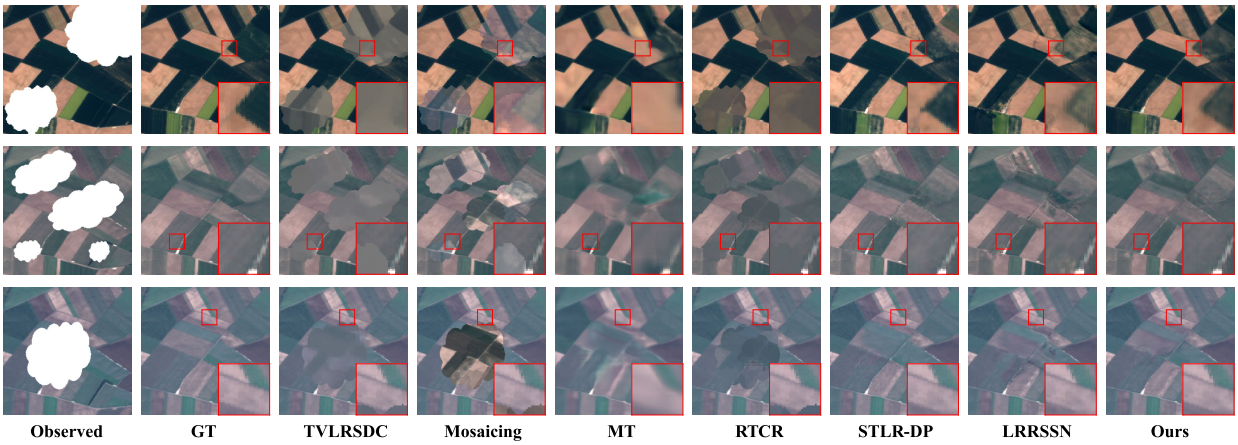


Fig. 4. Visual reconstruction results of different methods for the farmland dataset under four spectral bands. From top to bottom are the cloud removal results for time nodes 1, 2, and 3.

c) Village dataset: In the village dataset, we evaluate the impact of the number of time nodes in cloud-contaminated MRSIs. The dimension of the test images is $256 \times 256 \times 4 \times t$, with the number of time nodes $t = 3, 6, 9$, respectively. Moreover, cloud coverages of 30%–40% are applied to the first two-thirds of the time nodes.

d) Port dataset: Higher spatial resolutions can capture more image details, while lower resolutions may cause clouds to overlap with surface features, making it difficult to accurately recover the image. To investigate this, we consider cloud-contaminated data at three different spatial resolutions: $512 \times 512 \times 6 \times 6$ (original), $256 \times 256 \times 6 \times 6$, and $128 \times 128 \times 6 \times 6$. The lower resolution datasets are obtained from the original data via average pooling.

To quantitatively evaluate the performance of cloud removal results, three commonly used metrics are employed: peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM) [48], and correlation coefficient (CC) [49]. For these three quantitative metrics, higher values indicate better image reconstruction results. We fix $\lambda_1 = 0.005$, $\lambda_2 = 0.5$, $\rho_1 = 0.0001$, and $\rho_2 = 0.05$ in all simulated experiments, indicating the robustness of the proposed method.

2) Results on Farmland Dataset: Fig. 4 shows the visual results on the farmland dataset with four spectral bands. Model-driven methods, such as TVLRSDC and RTCR, exhibit significant blurriness in cloud-covered areas. The inpainting-based method mosaicing averages the clean regions of the time nodes corresponding to cloud locations, which results in noticeable discrepancies, particularly when the time intervals between nodes are large or surface changes are significant. Although the pure network method, MT, yields better visuals than the previous methods, it fails to recover many details, resulting in extensive blurring. STLR-DP and LRRSSN show improvements over other methods, but artifacts still appear in the cloud boundaries. In contrast, the proposed method achieves the best reconstruction result, recovering more details and maintaining spectral consistency. Table I presents the quantitative metrics of different methods on the farmland dataset. We can clearly observe that the proposed method achieves the best result across all metrics for varying spectral band counts, which proves the superiority of our method in reconstructing cloud-contaminated regions. In particular, compared to the model-driven methods, our approach significantly improves the PSNR by 7–8 dB, demonstrating the effectiveness of incorporating data-driven deep networks.

3) Results on Mountain Dataset: As shown in Fig. 5, when cloud coverage reaches 45%–60%, the overlapping cloud-covered areas from different time nodes lead to unsatisfactory recovery results for most methods. Both mosaicing and TVLRSDC exhibit varying degrees of spectral distortion, while MT and RTCR significantly compromise the color fidelity of the pseudo-color images and lose spatial details. Although STLR-DP and LRRSSN recover details better, they

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TABLE II
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE MOUNTAIN DATASET WITH VARYING CLOUD COVERAGES

Cloud coverage	Index	Observed	TVLRSDC	Mosaicing	MT	RTCR	STLR-DP	LRRSSN	Ours
15% - 30%	PSNR	13.066	43.227	37.237	33.590	44.471	48.957	50.370	**52.556**
	SSIM	0.895	0.990	0.980	0.882	0.989	0.995	0.997	**0.997**
	CC	0.247	0.994	0.968	0.944	0.995	0.998	0.999	**0.999**
30% - 45%	PSNR	8.134	33.826	32.142	28.326	38.107	40.482	41.746	**45.482**
	SSIM	0.727	0.947	0.939	0.831	0.964	0.981	0.985	**0.989**
	CC	0.147	0.949	0.918	0.820	0.980	0.989	0.991	**0.995**
45% -60%	PSNR	5.798	27.007	27.331	25.429	30.420	33.671	32.120	**36.872**
	SSIM	0.571	0.859	0.885	0.804	0.888	0.950	0.916	**0.971**
	CC	0.116	0.789	0.858	0.708	0.888	0.942	0.927	**0.976**

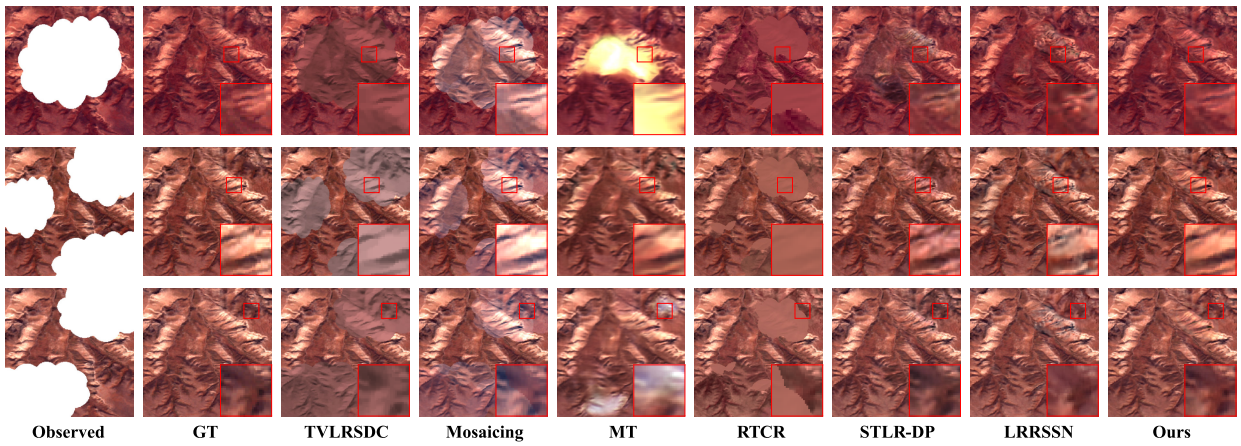


Fig. 5. Visual reconstruction results of different methods for the mountain dataset under cloud coverage 45%–60%. From top to bottom are the cloud removal results for time nodes 1, 2, and 3.

TABLE III
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS IN THE VILLAGE DATASET WITH VARYING NUMBERS OF TIME NODES

Time node	Index	Observed	TVLRSDC	Mosaicing	MT	RTCR	STLR-DP	LRRSSN	Ours
3	PSNR	6.721	29.171	28.438	29.588	29.298	30.281	28.009	**32.163**
	SSIM	0.637	0.879	0.913	0.783	0.870	0.903	0.866	**0.934**
	CC	0.108	0.870	0.884	0.881	0.875	0.923	0.846	**0.939**
6	PSNR	7.541	28.800	29.278	26.409	31.650	29.420	32.032	**34.971**
	SSIM	0.683	0.892	0.907	0.760	0.921	0.926	0.917	**0.951**
	CC	0.167	0.899	0.911	0.817	0.945	0.922	0.944	**0.973**
9	PSNR	7.291	29.565	30.627	27.385	32.727	31.541	33.699	**35.581**
	SSIM	0.676	0.902	0.915	0.783	0.925	0.941	0.937	**0.954**
	CC	0.149	0.910	0.930	0.829	0.957	0.941	0.965	**0.977**

still generate black artifacts in the first time node. In contrast, the proposed method maintains relatively excellent performance and spectral fidelity. Table II lists the quantitative metrics for different methods. As cloud coverage increases, all methods experience a decline in metrics, but our method consistently outperforms all others. In scenarios with high cloud coverage, the proposed method significantly outperforms STLR-DP and LRRSSN, highlighting the advantage of incorporating a self-supervised network to capture the local deep features.

4) Results on Village Dataset: Fig. 6 presents the reconstruction results for village dataset with six-time nodes. The results show that both TVLRSDC and mosaicing suffer from significant spectral distortion, while RTCR exhibits jagged artifacts at the cloud boundary in the fourth-time node. The MT method displays noticeable white artifacts in the cloud-covered regions of the first two time nodes, attributed to the limited generalization ability of the pure DIP framework for complex backgrounds. In contrast, the proposed method produces results closest to the ground truth. The quantitative results of different methods on the village dataset are presented in Table III. It is evident that the proposed method consistently outperforms the other methods in all cases.

5) Results on Port Dataset: Fig. 7 shows the reconstruction results for the port dataset with a spatial resolution of 256 × 256. TVLRSDC and MT exhibit significant blurriness in the cloud-covered areas, while mosaicing introduces noticeable spectral distortion at the second time node. The RTCR result shows a gray block due to cloud overlap across multiple time nodes in the region. In contrast, the

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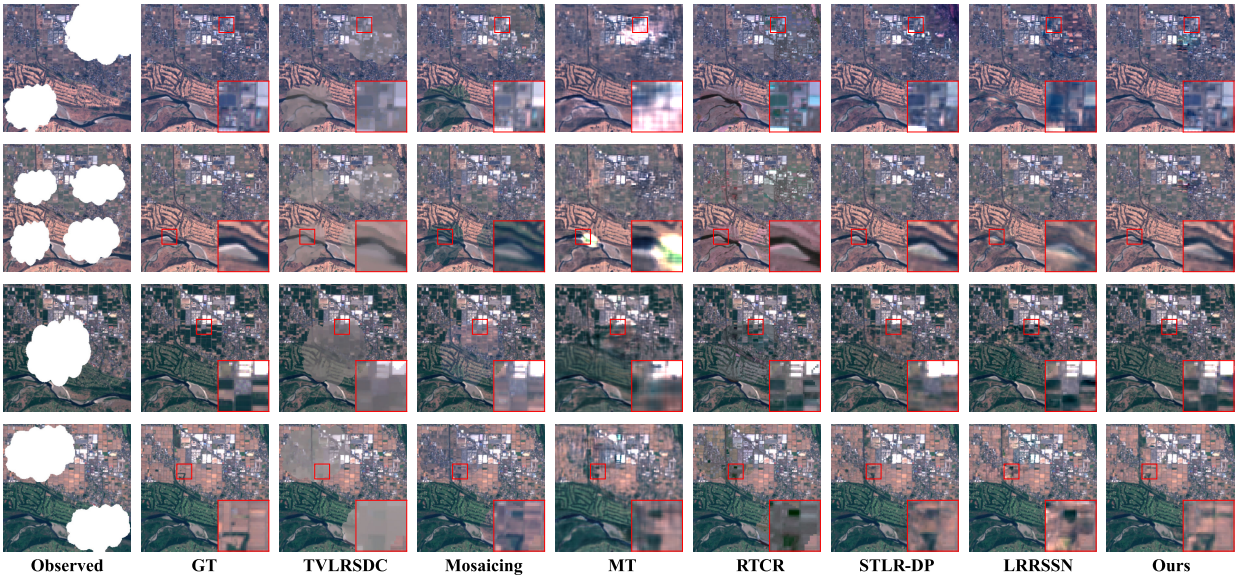


Fig. 6. Visual reconstruction results of different methods for the village dataset under six-time nodes. From top to bottom are the cloud removal results for time nodes 1, 2, 3, and 4.

TABLE IV
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS IN THE PORT DATASET WITH VARYING SPATIAL RESOLUTIONS

Spatial resolution	Index	Observed	TVLRSDC	Mosaicing	MT	RTCR	STLR-DP	LRRSSN	Ours
128 × 128	PSNR	7.486	36.271	31.855	31.450	35.836	27.104	37.973	**40.559**
	SSIM	0.637	0.979	0.951	0.867	0.977	0.920	0.984	**0.989**
	CC	0.338	0.987	0.958	0.961	0.987	0.977	0.991	**0.995**
256 × 256	PSNR	6.674	38.421	35.756	31.810	38.550	39.727	40.443	**43.054**
	SSIM	0.687	0.969	0.959	0.794	0.978	0.981	0.986	**0.989**
	CC	0.252	0.982	0.953	0.913	0.983	0.981	0.988	**0.993**
512 × 512	PSNR	6.171	41.154	43.328	32.897	41.130	37.722	44.099	**45.185**
	SSIM	0.708	0.970	0.984	0.776	0.979	0.963	0.988	**0.989**
	CC	0.197	0.976	0.986	0.828	0.977	0.988	0.986	**0.989**

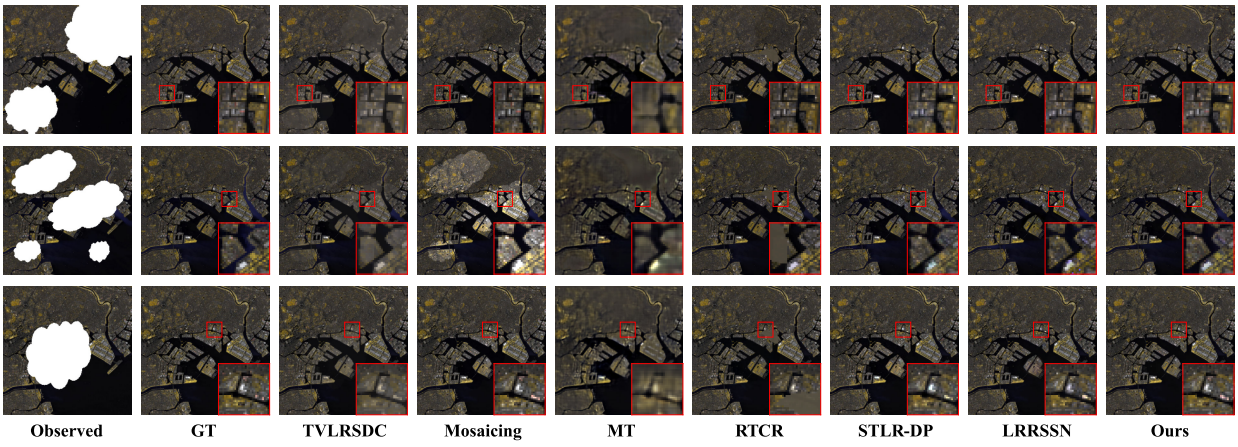


Fig. 7. Visual reconstruction results of different methods for the port dataset under spatial resolution 256 × 256. From top to bottom are the cloud removal results for time nodes 1, 2, and 3.

proposed method achieves superior reconstruction results, further demonstrating the advantage of the proposed joint-driven framework. Table IV lists the quantitative evaluation results for the port dataset. In high-resolution images, most methods achieve satisfactory results due to the detailed information provided. However, in low-resolution scenarios, the reconstruction results of most methods significantly degrade, indicating their difficulty in distinguishing the details of the image and the clouds. In contrast, the proposed method achieves optimal results at both resolutions, demonstrating that incorporating data-driven deep methods can effectively learn the image features.

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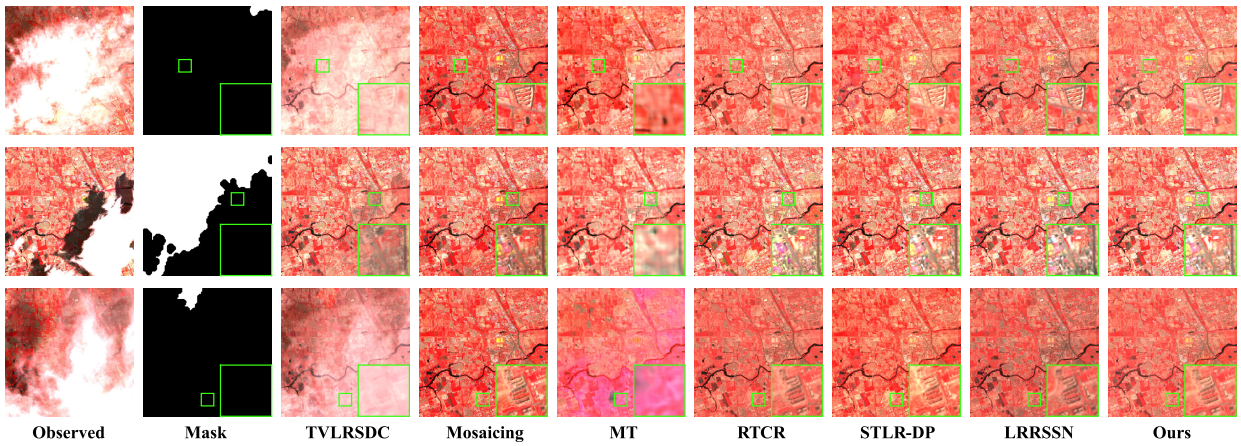


Fig. 8. Visual reconstruction results of different methods on the real SPOT-5 dataset. From top to bottom are the cloud removal results for time nodes 1, 2, and 3.

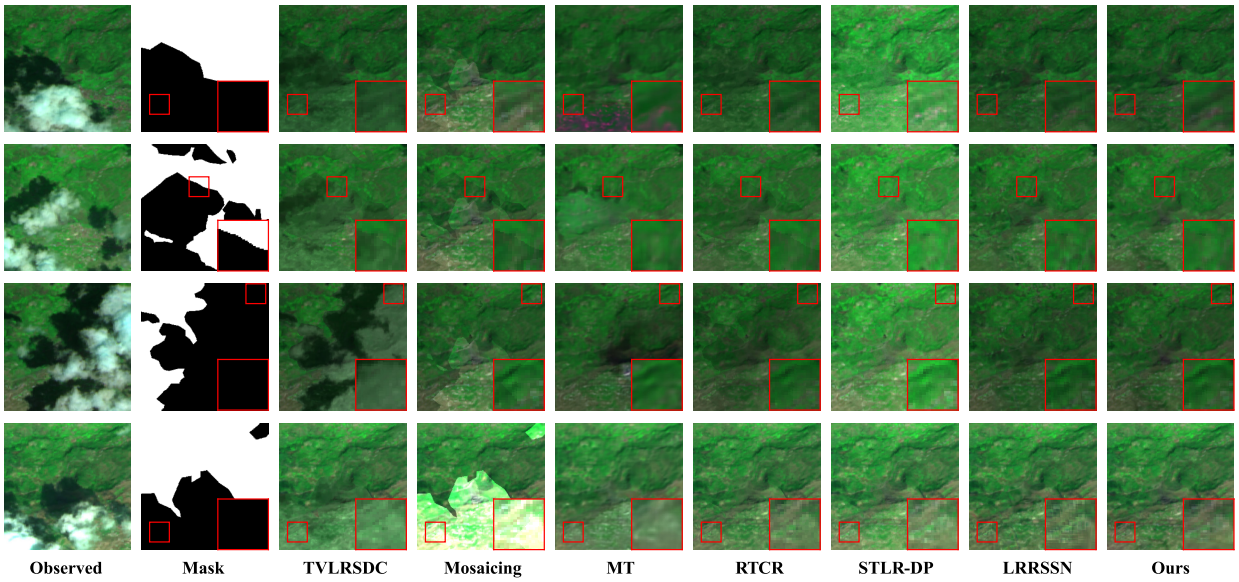


Fig. 9. Visual reconstruction results of different methods on the real SEN12MS-CR-TS dataset. From top to bottom are the cloud removal results for time nodes 1, 2, 3, and 5.

B. Real Experiment

1) Real Experiment Setting: To further validate the effectiveness of the proposed method in real-world scenarios and under complex cloud conditions, we test it on two complex real-world datasets. The first dataset is the MRSIs data captured by the SPOT-5 satellite sensor over Beijing in 2015. It consists of four spectral bands: red, green, blue, and near-infrared, with a spatial resolution of 10 m. For testing, we select six time nodes and crop the data to a size of 512×512 for easier processing. The first three time nodes are significantly affected by extensive cloud coverage. The second dataset is derived from the SEN12MS-CR-TS dataset, with a shape of $256 \times 256 \times 13 \times 5$, where only the fourth time node is cloud-free, while the other four time nodes suffer from severe contamination due to thick clouds and shadows, which makes recovery more challenging. In the second column of Figs. 8 and 9, the masks with overcoverage are provided for the semi-blind and nonblind comparison methods. Shadow areas are also included in these masks to enhance the overall restoration quality. The parameters $\lambda_1 = 0.05$, $\lambda_2 = 0.5$, $\rho_1 = 0.0001$, and $\rho_2 = 0.05$ are set in the real experiments.

2) Results on SPOT-5 Dataset: As shown in Fig. 8, the real SPOT-5 dataset is significantly affected by cloud and shadow, particularly at the first and third time nodes. It can be observed that TVLRSDC fails to completely remove the clouds, only reducing their thickness. Although mosaicing produces relatively good results, it performs noticeable hue differences in the restored areas beneath the clouds, as these are generated by averaging pixels from cloud-free time nodes. In comparison, STLR-DP and LRRSSN achieve relatively ideal visual effects, successfully reconstructing most of the missing information beneath the clouds, but some details are smoothed or distorted, as seen in the enlarged view. Our method fully reconstructs the cloud-affected areas while preserving the original details in the image, maintaining a high degree of spectral fidelity with the cloud-free areas. This effectiveness is attributed to our method's ability to effectively capture the physical prior information of the image and cloud

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TABLE V
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS IN THE REAL DATASET

Dataset	Index	TVLRSDC	Mosaicing	MT	RTCR	STLR-DP	LRRSSN	Ours
SPOT-5	BRISQUE ↓	13.573	10.179	48.09	11.187	12.269	<u>9.993</u>	**9.345**
	metric Q ↑	0.646	0.737	0.652	0.749	<u>0.753</u>	0.752	**0.755**
SEN12MS-CR-TS	BRISQUE ↓	27.341	23.177	38.346	20.660	**18.677**	23.469	<u>19.496</u>
	metric Q ↑	0.578	0.579	0.596	<u>0.620</u>	0.605	0.602	**0.626**

TABLE VI
PSNR RESULTS OF ABLATION STUDY ON THE FARMLAND AND MOUNTAIN DATASETS

Dataset	w/o LR	w/o GS	w/o ADG	w/o U-Net	Data-driven	Model-driven	Ours
Farmland	35.566	35.747	27.153	34.017	36.557	27.150	**37.267**
Mountain	43.602	44.111	40.103	45.154	44.885	40.094	**45.482**

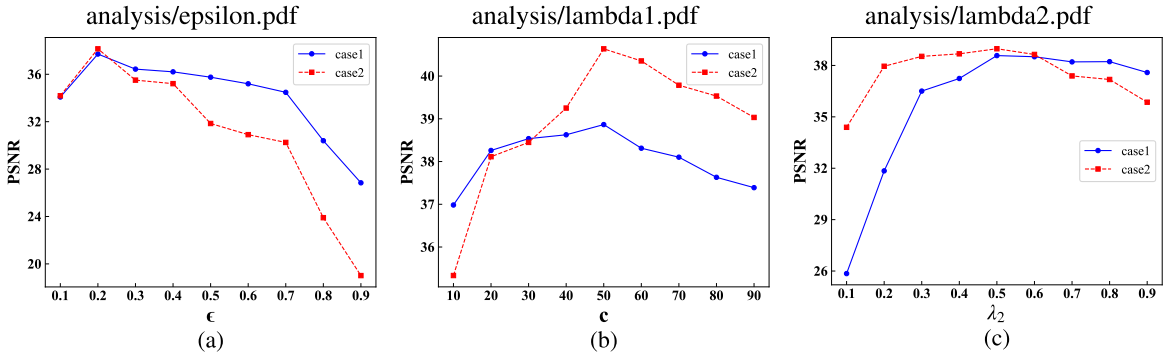


Fig. 10. Sensitivity analysis of parameters ϵ, λ_1, and λ_2 under the case of farmland dataset with four spectral bands (Case 1) and the case of mountain dataset with cloud coverage of 45%–60% (Case 2). (a) Mask threshold ϵ. (b) Tuning parameter c of λ_1. (c) Regularization parameter λ_2.

components, complemented by deep prior information for detail recovery.

3) Results on SEN12MS-CR-TS Dataset: Fig. 9 displays the visual reconstruction results of various methods on the SEN12MS-CR-TS dataset. The results reveal that TVLRSDC and MT share similar flaws seen in the SPOT-5 dataset: incomplete cloud removal and pixel blurring. Due to stark color variations across time nodes, mosaicing becomes unsuitable, exhibiting clear color inconsistencies. STLR-DP offers better reconstruction but introduces a tonal mismatch with the observed image. In contrast, our method demonstrates promising visual performance.

4) Quantitative Analysis: In the real experiments, two no-reference image quality assessment metrics: BRISQUE [50] and metric Q [51], are adopted to evaluate the quality of the reconstructed images. In Table V, the best values are highlighted in bold, while the second-best are underlined. As shown, the proposed method achieves superior performance on both datasets, further demonstrating its effectiveness. The next best results are obtained by two hybrid-based methods, which is consistent with the visual observations.

C. Ablation Study and Discussion

1) Ablation Study: To evaluate the contribution of each module in our method, we conduct ablation experiments on two scenes. The proposed method employs a model-driven low-rank group sparse decomposition module to capture global structural features of the image and cloud components, and data-driven dual self-supervised deep networks to learn the local deep features. We perform experiments by sequentially disabling each module: low-rank regularization (w/o LR), group sparse regularization (w/o GS), ADG (w/o ADG), and U-Net (w/o U-Net). In addition, the proposed method combines the advantages of both model-driven and data-driven approaches. To further validate its effectiveness, we separately remove the model-driven regularization and the data-driven self-supervised network, referring to them as model-driven and data-driven, respectively. Table VI reports the PSNR results of ablation studies on the farmland and mountain datasets. The proposed method outperforms all variants, confirming that each module plays a vital role. The results also show that model-driven and data-driven components are complementary, and their combination of global structural and local deep features significantly boosts performance.

2) Analysis of Mask Threshold: To evaluate the impact of selecting different mask thresholds ϵ on experimental performance, we conduct sensitivity analysis experiments for ϵ on a farmland dataset with four bands (case1) and a mountain dataset with cloud coverage ranging from 45%–60% (case2). Fig. 10(a) illustrates the quantitative recovery results under different ϵ values. The results demonstrate that setting $\epsilon = 0.2$ achieves optimal performance. However, when ϵ exceeds 0.2, less prominent cloud regions are neglected, resulting in a

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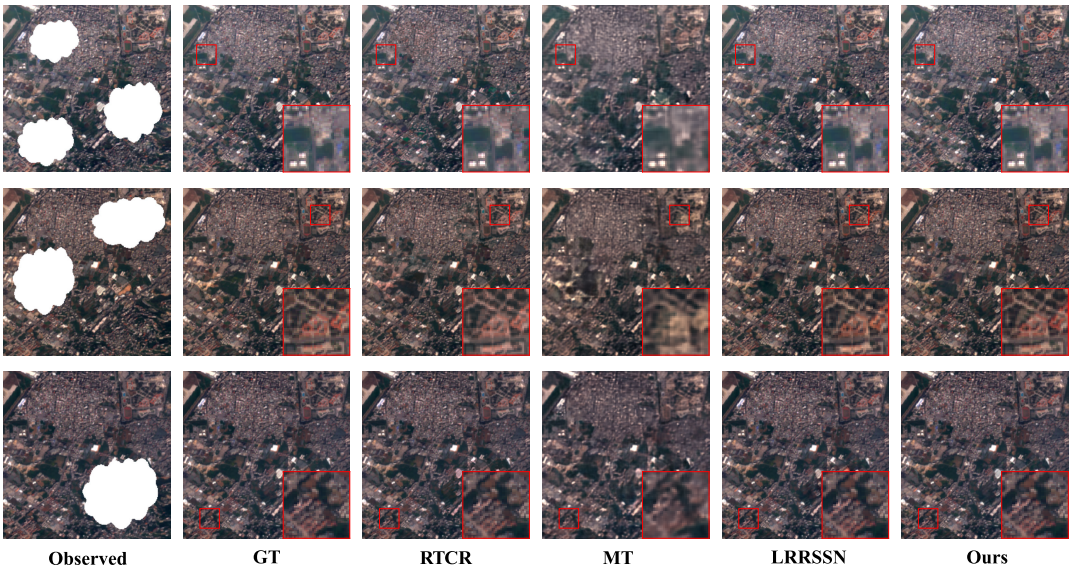


Fig. 11. Visual reconstruction results of different methods for the city dataset under Case A. From top to bottom are the cloud removal results for time nodes 1, 2, and 3.

TABLE VII
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE CITY DATASET WITH VARYING CLOUD COVERAGES

Case	Index	Observed	RTCR	MT	LRRSSN	Ours
Case A	PSNR	10.012	35.451	27.304	35.143	**37.706**
	SSIM	0.782	0.977	0.760	0.973	**0.984**
	CC	0.291	0.984	0.895	0.981	**0.990**
Case B	PSNR	8.427	31.764	26.960	30.584	**32.071**
	SSIM	0.705	0.946	0.756	0.935	**0.958**
	CC	0.250	0.960	0.885	0.946	**0.964**
Case C	PSNR	6.551	29.251	26.092	29.308	**30.064**
	SSIM	0.582	0.898	0.730	0.912	**0.929**
	CC	0.131	0.935	0.859	0.937	**0.944**

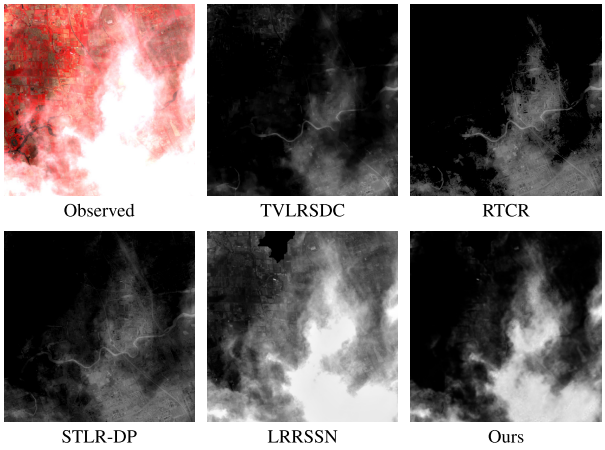


Fig. 12. Cloud component estimated by different methods at the third time node on the real SPOT-5 dataset.

substantial decline in PSNR values. Consequently, we adopted $\epsilon = 0.2$ for all experiment.

3) Analysis of Regularization Parameter: The proposed method involves two regularization parameters, λ_1 and λ_2, which, respectively, control the influence of the low-rank regularizer for the image component and the group sparsity regularizer for the cloud component. To evaluate the impact of these parameters on model performance, we conduct experiments on the same two datasets used in the mask threshold analysis. Specifically, we define $\lambda_1 = c\rho_1$, where c is a tuning parameter that controls the scaling factor between the two. Fig. 10(b) and (c) illustrates the variation trends of PSNR values with respect to the tuning parameters c and λ_2, respectively. As shown in the figures, the optimal performance across different scenarios is achieved when $c = 50$ and $\lambda_2 = 0.5$. Consequently, these values were adopted for all experiments.

4) Performance in Complex Scenarios: To validate the robustness of the proposed method in complex scenarios, we select a complex urban scene from the SEN12MS-CR-TS dataset and introduce three levels of cloud coverage (consistent with the settings of the mountain dataset) as Cases A, B, and C. We compare our method with SOTA approaches from three different categories: the model-based method RTCR, the deep learning-based method MT, and the hybrid method LRRSSN. Fig. 11 shows the recovery results of different methods under Case A, while Table VII presents the corresponding quantitative metrics. It can be observed that the proposed method achieves the best performance in both texture detail restoration and quantitative evaluation, further demonstrating its robustness in complex scenarios.

5) Analysis of Cloud Extraction Performance: To validate the effectiveness of the proposed method in cloud component extraction, Fig. 12 shows the extraction results of different approaches on the real SPOT-5 dataset. Since some methods do not explicitly extract cloud components, their results are not included in the figure. As shown, the first three methods extract only an extremely thin cloud layer, while LRRSSN suffers from overextraction, misclassifying ground texture details as part of the cloud components. These findings indicate that relying solely on physical priors is insufficient to accurately

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TABLE VIII
MODEL EFFICIENCY ANALYSIS ON SEN12MS-CR-TS DATASET

Model	Parameters/M	FLOPs/G	Run-times/s
MT	2.44	**3.42**	1875.33
STLR-DP	2.26	20.81	1325.29
LRRSSN	**0.08**	4.62	1256.97
Ours	14.05	22.61	**1183.69**

capture cloud structures. In contrast, the proposed method successfully extracts cloud components with complex structures, further demonstrating the effectiveness of incorporating neural networks for cloud extraction.

6) Model Complexity Analysis: To evaluate the computational efficiency and model complexity of the proposed method, Table VIII reports the number of model parameters, floating-point operations (FLOPs), and processing time for different approaches on the real-world SEN12MS-CR-TS dataset. Model-based or nondeep learning inpainting methods are excluded, as they lack learnable parameters and their computational complexity cannot be reliably measured by standard FLOPs metrics. As shown in the table, LRRSSN and MT have a small number of parameters, yet their runtime is relatively long. This is mainly because both methods are based on lightweight networks but require a large number of iterations. In contrast, the proposed method achieves a good trade-off between time efficiency and restoration performance.

V. CONCLUSION

In this article, we have proposed a novel cloud removal method to address the issue of thick cloud coverage in MRSIs, which does not require any external training data or pretrained models and is capable of adaptively applying to different scenes and factors. The proposed method equally treats both image and cloud components and effectively integrates data-driven and model-driven approaches to fuse their respective global structural features and local deep features. In the joint-driven framework, the image and cloud components can be estimated iteratively, in which the iterative estimation manner would benefit from each other. The model-driven low-rank group sparsity decomposition framework effectively captures the global structural features of the image and cloud components, allowing for their discriminative separation. In contrast, the data-driven self-supervised network adaptively learns the local deep features of both the image and cloud components, facilitating the identification of fine local details. The proposed method tests on various simulated and real thick cloud images and achieves better cloud removal performance than compared methods, in terms of different scenes and factors.

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